

IONIA COUNTY AP, MI, USA

WMO: 720661

Lat: 42.938N

Lon: 85.061W

Elev: 249

StdP: 98.37

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-13.5	-21.1	0.6	-15.2	-17.9	0.8	-12.1	12.0	-2.1	11.0	-1.4	2.3	290	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.4	22.0	30.9	21.5	29.1	20.8	23.9	29.3	23.1	28.2	22.3	27.1	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.6	26.4	21.7	16.9	25.5	20.9	16.1	24.9	72.9	29.7	69.7	28.2	66.8	26.9	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.1	8.2	-20.2	34.6	3.5	2.1	-22.7	36.1	-24.8	37.3	-26.7	38.5	-29.3	40.0		
			-20.6	25.4	3.5	0.7	-23.1	25.9	-25.1	26.3	-27.0	26.8	-29.6	27.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.7	-2.9	2.2	8.1	15.6	20.0	23.1	21.2	17.7	11.0	4.5	-0.4		
	DBStd	10.43	5.80	5.65	5.91	4.86	4.99	3.52	2.97	2.74	4.20	4.64	4.98	4.59		
	HDD10.0	1687	423	362	254	92	9	0	0	0	1	44	178	323		
	HDD18.3	3523	682	595	501	309	116	21	2	7	63	230	416	582		
	CDD10.0	1603	0	0	12	33	183	299	406	348	232	77	12	1		
	CDD18.3	398	0	0	1	1	31	71	149	97	44	4	0	0		
	CDH23.3	3578	0	0	7	22	323	652	1443	740	349	41	0	0		
	CDH26.7	1118	0	0	1	2	96	197	526	191	102	4	0	0		
(14) Wind	WSAvg	3.7	4.2	4.2	4.0	4.4	3.8	3.4	2.7	2.5	2.9	3.7	4.1	3.9		
(15) Precipitation	PrecAvg	860	59	50	49	78	103	89	81	74	82	77	70	56		
	PrecMax	1052	116	116	100	161	250	194	139	164	224	153	176	138		
	PrecMin	696	17	19	10	22	43	29	21	23	13	2	20	22		
	PrecStd	113	31	23	25	33	55	45	33	38	49	42	43	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.9	25.2	26.3	32.1	33.3	36.0	32.4	32.4	27.2	21.3	14.9		
		MCWB	11.8	10.4	16.5	16.0	20.4	22.5	23.1	21.9	21.0	17.4	14.5	12.4		
	2%	DB	8.9	10.4	18.0	22.5	29.1	31.0	32.7	30.2	29.8	24.0	17.1	11.1		
		MCWB	6.7	7.2	13.0	14.5	19.4	20.7	22.7	21.6	22.3	16.9	11.9	9.3		
	5%	DB	5.2	6.8	14.7	19.0	27.3	28.9	31.2	28.8	27.2	21.6	14.8	7.9		
		MCWB	3.3	4.3	10.6	12.1	18.3	20.2	22.0	21.1	20.1	16.6	11.0	6.6		
	10%	DB	3.3	4.0	11.2	16.8	24.6	27.2	29.5	27.4	25.0	18.9	12.4	5.9		
		MCWB	2.0	2.1	7.8	10.8	17.0	19.2	21.3	20.3	19.1	14.8	9.0	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.4	17.5	18.2	22.1	24.3	25.5	24.1	23.9	20.6	15.9	13.6		
		MCDB	12.7	13.3	24.6	23.6	27.8	30.0	31.6	28.5	29.0	25.1	19.4	14.8		
	2%	WB	6.9	7.7	14.6	15.2	20.8	23.0	24.3	23.2	22.8	18.8	13.3	9.3		
		MCDB	8.3	10.5	17.5	19.7	27.1	28.4	30.1	28.0	27.5	22.5	16.7	10.6		
	5%	WB	3.9	4.6	11.6	13.3	19.7	21.7	23.4	22.4	21.7	17.0	11.0	7.0		
		MCDB	5.5	6.5	14.0	17.9	25.4	26.9	28.7	26.8	25.3	19.8	14.2	7.8		
	10%	WB	2.3	2.4	7.8	11.4	18.2	20.5	22.6	21.5	20.4	15.3	9.3	4.4		
		MCDB	3.3	4.1	10.5	15.9	22.9	25.2	27.6	25.3	23.7	19.3	12.2	5.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.6	9.1	10.6	10.9	10.9	11.4	11.0	11.1	9.8	8.1	5.8		
		MCDBR	9.2	9.6	13.5	15.0	13.5	12.6	13.0	12.4	12.7	13.8	11.8	8.9		
		MCWBR	8.0	7.7	8.8	8.4	6.1	5.7	5.2	5.4	5.5	7.4	7.9	7.6		
		MCDBR	8.6	9.5	12.0	12.8	12.5	10.6	10.7	10.3	10.8	11.5	11.2	8.5		
	5% WB	MCWBR	7.7	8.1	8.8	8.7	6.3	6.0	5.2	5.3	5.4	8.1	8.6	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.347	0.337	0.378	0.410	0.431	0.420	0.418	0.380	0.344	0.313	0.294		
	taud		2.349	2.168	2.365	2.276	2.236	2.215	2.249	2.253	2.345	2.445	2.477	2.406		
	Ebn at Noon		839	833	905	888	865	844	849	839	847	839	818	808		
	Edh at Noon		81	115	108	127	137	140	134	129	109	87	72	71		
(44) All-Sky Solar Radiation	RadAvg		1.36	2.23	3.56	4.59	5.37	6.05	6.14	5.25	4.22	2.65	1.61	1.10		
	RadStd		0.12	0.28	0.38	0.42	0.52	0.38	0.29	0.28	0.40	0.29	0.22	0.12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page